

CHE 201 HW 7

Due: 11.59pm on 04/05/2020

1. Refrigerant 134A enters a well-insulated nozzle at 200 lbf/in², 220 °F, with a velocity of 120 ft/s and exits at 20 lbf/in² with a velocity of 1500 ft/s. For steady state operation and neglecting potential energy effects:
 - a) Draw a schematic of the system and label the system boundary.
 - b) State the engineering model
 - c) Write the simplified mass rate and energy rate balance.
 - d) Determine the exit temperature in °F
 - e) Carefully sketch the process on a P-v diagram

2. Air expands through a turbine from 8 bar, 960 K to 1 bar, 450 K. The inlet velocity is small compared to the exit velocity of 90 m/s. The turbine operates at steady state and develops a power output of 2500 kW. Heat transfer between the turbine and its surroundings and potential energy effects are negligible. Modeling air as an ideal gas:
 - a) Draw a schematic of the system and label the system boundary.
 - b) State the engineering model
 - c) Write the simplified mass rate and energy rate balance.
 - d) Calculate the mass flow rate of the air, in kg/s, and the exit area, in m².
 - e) Carefully sketch the process on a T-v diagram

3. Water vapor enters a turbine operating at steady state at 500°C, 40 bar, with a velocity of 200 m/s, and expands adiabatically to the exit, where it is a saturated vapor at 0.8 bar, with a velocity of 150 m/s and a volumetric flow rate of 9.48 m³/s. Neglecting potential energy effects:
 - a) Draw a schematic of the system and label the system boundary.
 - b) Carefully sketch the process on a P-v diagram
 - c) State the engineering model
 - d) Write the simplified mass rate and energy rate balance.
 - e) Determine the power developed by the turbine, in kW.

4. Refrigerant 134A enters an air conditioner compressor at 4 bar, 20°C, and is compressed at steady state to 12 bar, 80 °C. The volumetric flow rate of the refrigerant entering is 4 m³/min. The work input to the compressor is 60 kJ per kg of refrigerant flowing.

Neglecting kinetic and potential energy effects:

- Draw a schematic of the system and label the system boundary.
 - Carefully sketch the process on a T-v diagram
 - State the engineering model
 - Write the simplified mass rate and energy rate balance.
 - Determine the heat transfer rate, in kW
5. A pump at steady state is drawing water at a temperature of 15 °C and a pressure of 1 bar from a reservoir and delivering it at a pressure of 3 bar to a storage tank that is 15 m above reservoir. The mass flow rate of the water is 1.5 kg/s. The water temperature remains nearly constant at 15 °C, and there is no significant change in kinetic energy from inlet to exit. Heat transfer between the pump and the surroundings is negligible.
- Draw a schematic of the system and label the system boundary.
 - State the engineering model
 - Write the simplified mass rate and energy rate balance.
 - Determine the power required by the pump, in kW. Let $g = 9.81 \text{ m/s}^2$
6. Ammonia enters the expansion valve of a refrigeration system at a pressure of 10 bar, and temperature of 24 °C, and exits a 1 bar. If the refrigerant undergoes a throttling process:
- Draw a schematic of the system and label the system boundary.
 - State the engineering model
 - Write the simplified mass rate and energy rate balance.
 - Determine the quality of the refrigerant exiting the expansion valve.
 - Carefully sketch the process on a T-v diagram